IT2525/IT2130 Cyber Forensic Technologies & Process
AY2026 Semester 2

Practical 1 – Storage File Systems

Learning Outome

To understand, examine and explain various industry standard file system formats used in different operating systems.

After this practical lab, you will able to:

1) Use AccessData FTK Imager to examine, locate and describe the evidence in

   - MBR
   - FAT file system

Task 1	**Examine, locate and explain the file system parameters and evidence in a given disk image file.**

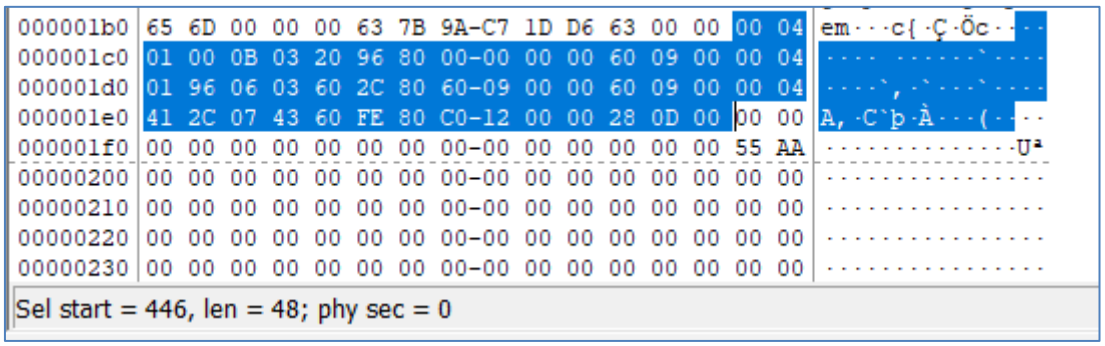
Instructions

1. Download a copy of the disk image file `123456A-FAT32.E01` from Politemall into the Windows forensic virtual machine.

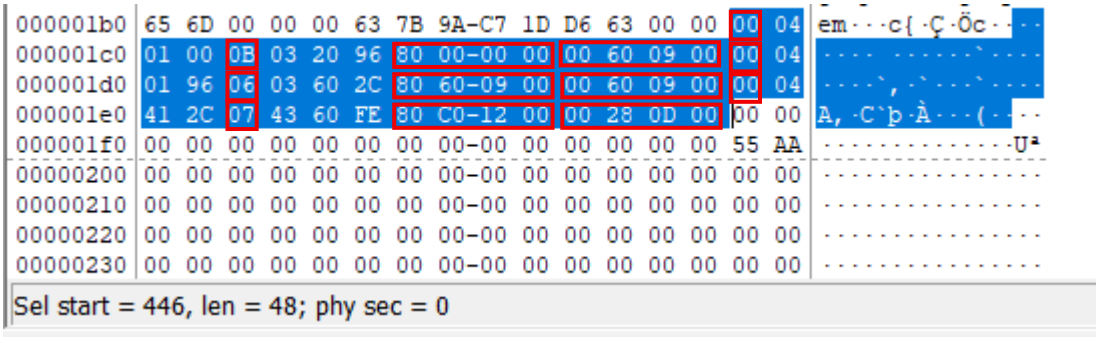
2. Use AccessData FTK Imager to examine the disk image file.

Question 1

a) Locate Master Boot Record (MBR) in physical sector 0, and take a screenshot of the partition table in the Master Boot Record (MBR)
(Note: Refer to the slide “Structure of a Generic MBR: in Week 2 lecture note”)



b) For each of the non-zero partition entries in the partition table, use rectangles to mark the following parameters.

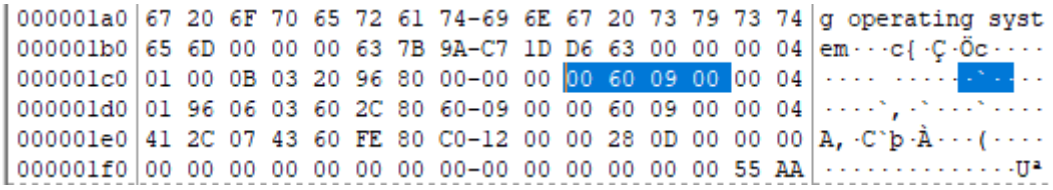


a. Active or inactive partition
Partition 1 byte offset 0x0 contains 0x00, means inactive partition

b. Partition type
Partition 1 byte offset 0x4 contains 0x0B, means partition type is FAT32 with CHS addressing

c. Starting sector of the partition
Partition 1 byte offset 0x8 to 0xB contains the number 0x00000080 (read in little endian byte order), means starting sector (1st sector) of the partition is 0x00000080.

d. Size of the partition in sectors
Partition 1 byte offset 0x8 to 0xB contains the number 0x00000080 (read in little endian way), means starting sector (1st sector) of the partition is 0x00000080.

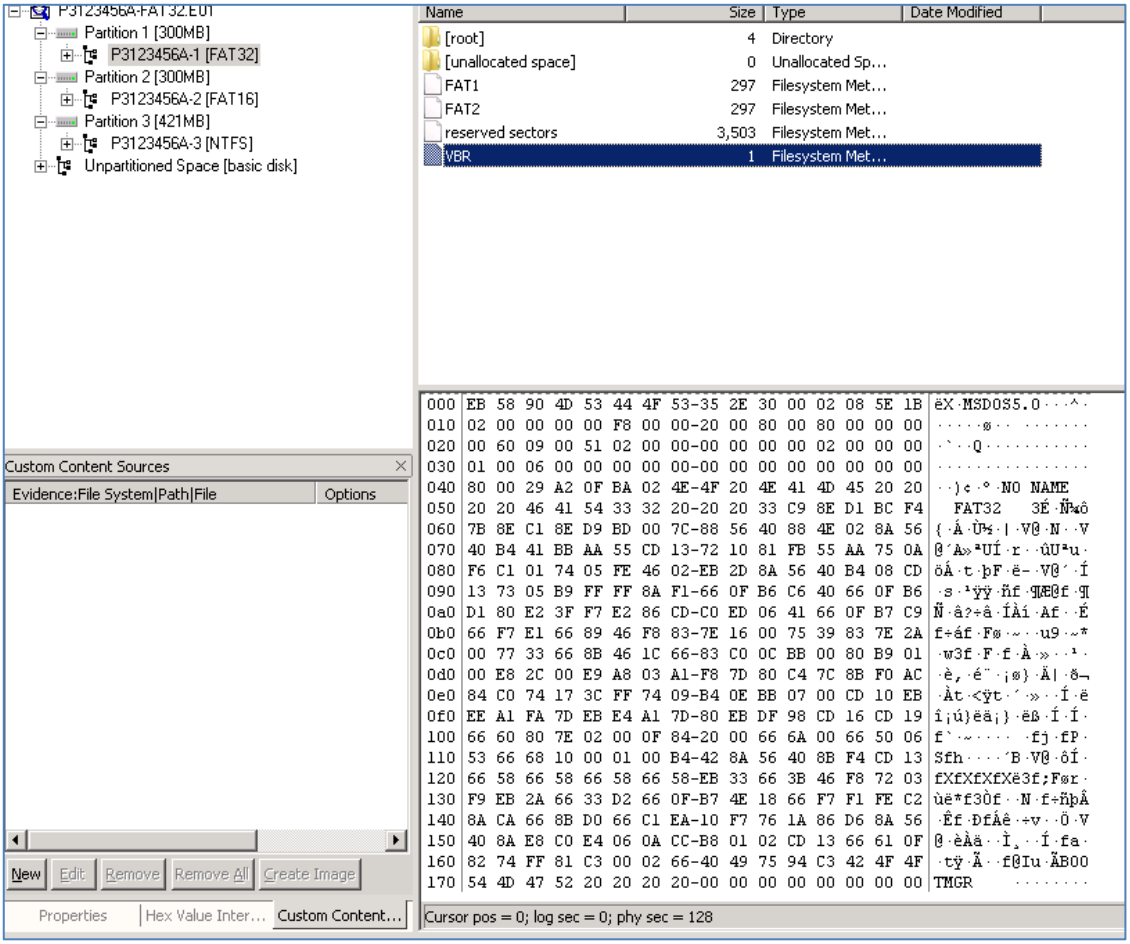


c) Examine the parameters in the partition table and fill the parameters in the partition table below.

Entry	Byte offset 0x0 Active or inactive	Partition Type (e.g. FAT32) Byte offset 0x04	Starting sector of the partition in decimal (Byte offset 0x8 to 0xB)	Size of the partition in sectors (in decimal) Byte offset 0xC to 0xF
1	0x00 inactive	0x0B FAT32 with CHS Addressing	0x00000080 or 128 in decimal	0x00096000 or 614,400 in decimal
2	0x00 inactive	0x06 FAT16B	0x00096080 or 614,528 in decimal	0x00096000 or 614,400 in decimal
3	0x00 inactive	0x07 NTFS, ExFAT	0x0012C080 or 1,228,928 in decimal	0x000D2800 or 862,208 in decimal

Refer to the file **FAT12 FAT16 and FAT32 file system formats** file in Politemall to examine the 3 file system components: VBR, FAT, and Root Folder in the **first FAT32 file volume** of the disk image file.

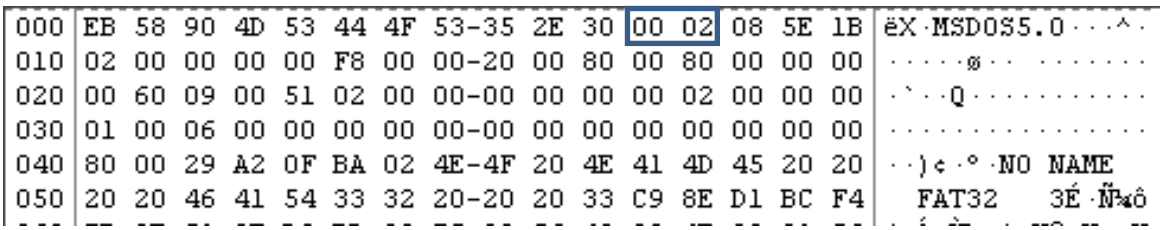
d) Click to select the Volume Boot Record (VBR) in the first simple volume of the disk



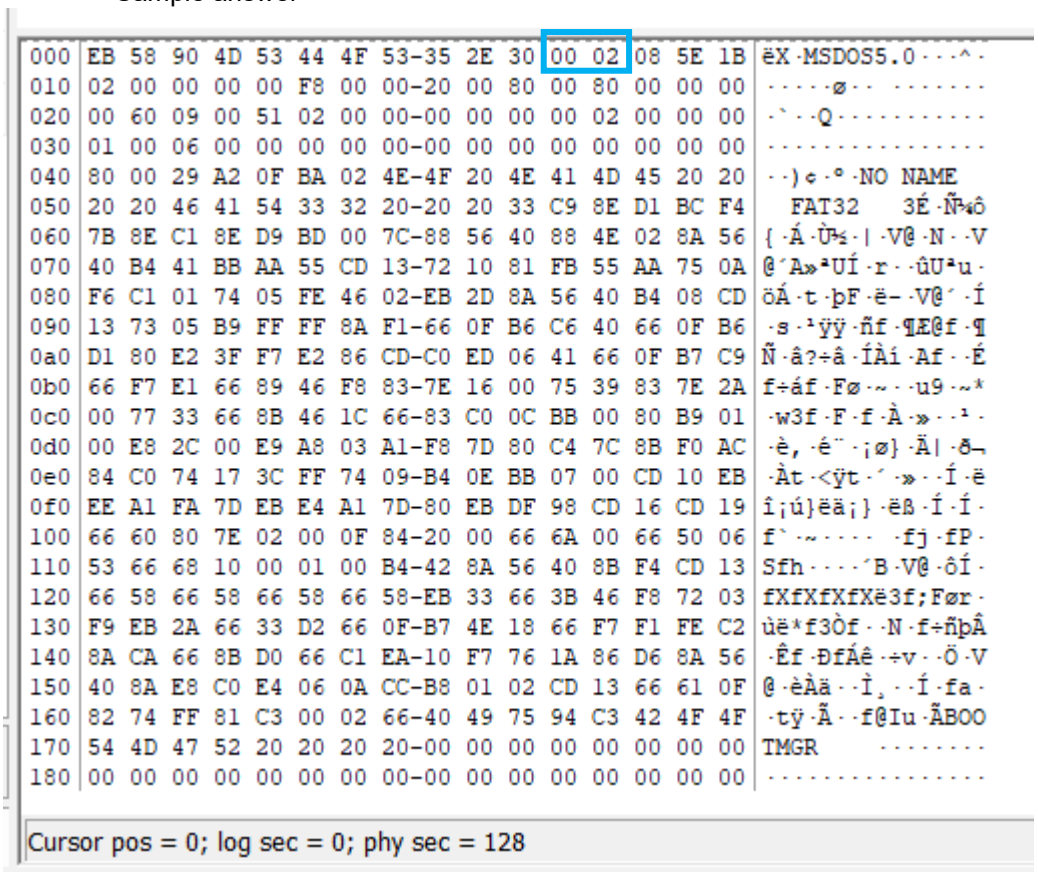
image

e) Capture the screen that shows the parameters of the first 90 bytes (offset 0x00 to 0x5A) of the VBR.

f) Refer to the **FAT32 File System Format**. What is the size (in bytes) of a sector in the file system? Mark the size of a sector parameter in the captured screen (similar to the image below)



Sample answer

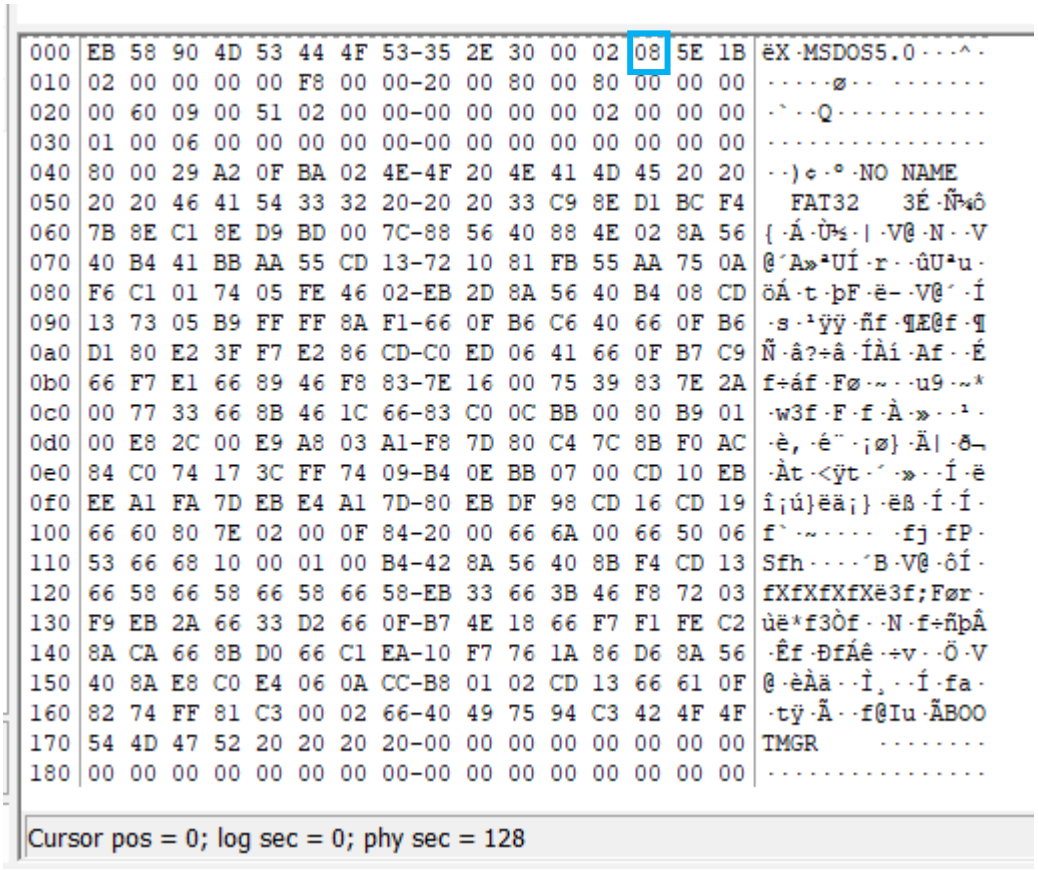


000	EB	58	90	4D	53	44	4F	53-35	2E	30	00	02	08	5E	1B
010	02	00	00	00	00	F8	00	00-20	00	80	00	80	00	00	00
020	00	60	09	00	51	02	00	00-00	00	00	00	02	00	00	00
030	01	00	06	00	00	00	00	00-00	00	00	00	00	00	00	00
040	80	00	29	A2	0F	BA	02	4E-4F	20	4E	41	4D	45	20	20
050	20	20	46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4
060	7B	8E	C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56
070	40	B4	41	BB	AA	55	CD	13-72	10	81	FB	55	AA	75	0A
080	F6	C1	01	74	05	FE	46	02-EB	2D	8A	56	40	B4	08	CD
090	13	73	05	B9	FF	FF	8A	F1-66	0F	B6	C6	40	66	0F	B6
0a0	D1	80	E2	3F	F7	E2	86	CD-C0	ED	06	41	66	0F	B7	C9
0b0	66	F7	E1	66	89	46	F8	83-7E	16	00	75	39	83	7E	2A
0c0	00	77	33	66	8B	46	1C	66-83	C0	0C	BB	00	80	B9	01
0d0	00	E8	2C	00	E9	A8	03	A1-F8	7D	80	C4	7C	8B	F0	AC
0e0	84	C0	74	17	3C	FF	74	09-B4	0E	BB	07	00	CD	10	EB
0f0	EE	A1	FA	7D	EB	E4	A1	7D-80	EB	DF	98	CD	16	CD	19
100	66	60	80	7E	02	00	0F	84-20	00	66	6A	00	66	50	06
110	53	66	68	10	00	01	00	B4-42	8A	56	40	8B	F4	CD	13
120	66	58	66	58	66	58	66	58-EB	33	66	3B	46	F8	72	03
130	F9	EB	2A	66	33	D2	66	0F-B7	4E	18	66	F7	F1	FE	C2
140	8A	CA	66	8B	D0	66	C1	EA-10	F7	76	1A	86	D6	8A	56
150	40	8A	E8	C0	E4	06	0A	CC-B8	01	02	CD	13	66	61	0F
160	82	74	FF	81	C3	00	02	66-40	49	75	94	C3	42	4F	4F
170	54	4D	47	52	20	20	20	20-00	00	00	00	00	00	00	00
180	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00

Number of bytes per sector is a WORD (2 bytes) number starting from byte offset 0xB. From the screenshot above, the number of bytes per sector is 0x0200, or 512 bytes in decimal.

g) What is the number of sectors in a cluster in the file system?

Number of sectors in a cluster is a one-byte number starting from byte offset 0xD. From the screenshot above, the number of sectors in a cluster is 0x08, or 8 sectors in decimal.

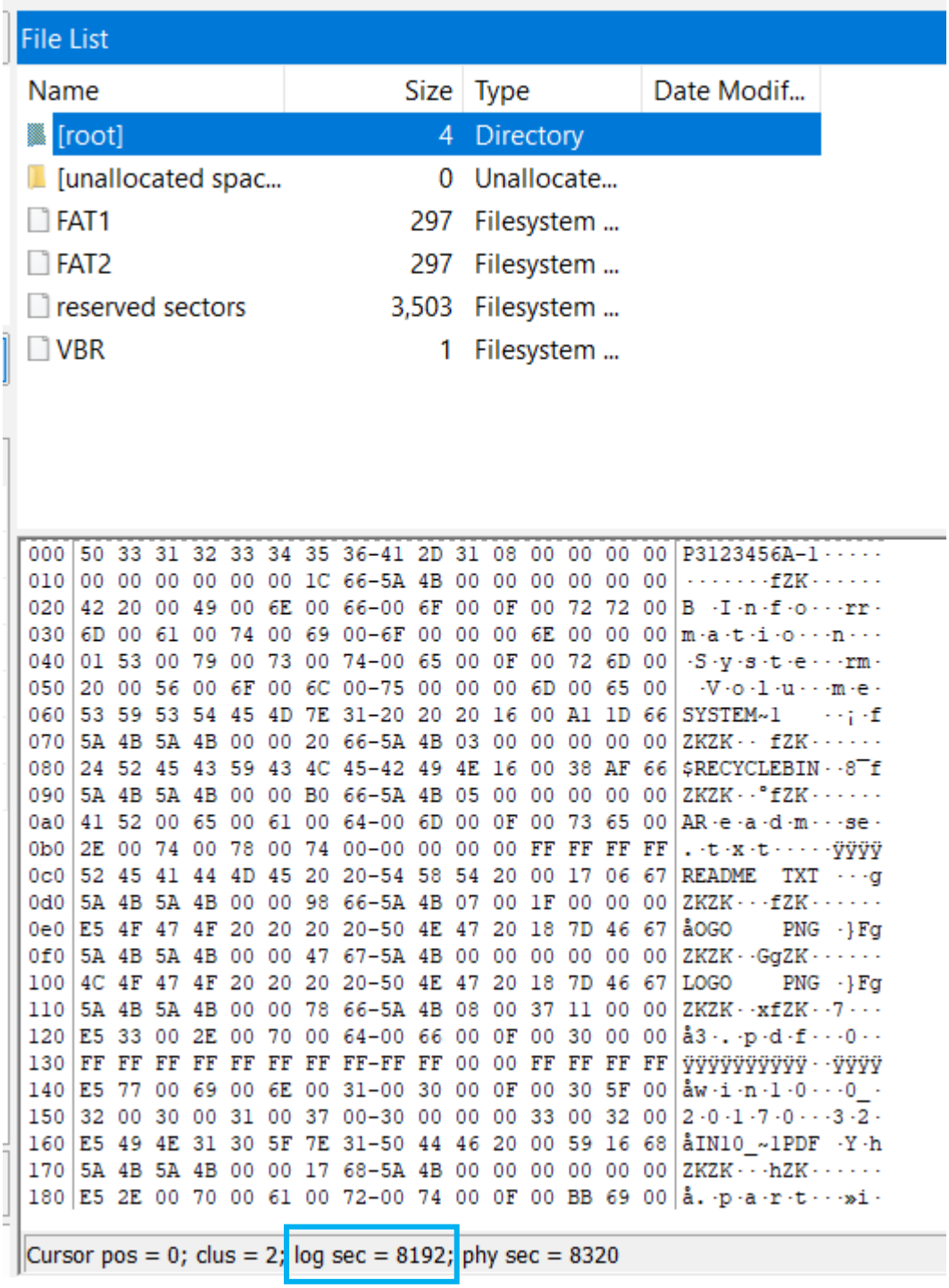


000	EB	58	90	4D	53	44	4F	53-35	2E	30	00	02	08	5E	1B
010	02	00	00	00	00	F8	00	00-20	00	80	00	80	00	00	00
020	00	60	09	00	51	02	00	00-00	00	00	00	02	00	00	00
030	01	00	06	00	00	00	00	00-00	00	00	00	00	00	00	00
040	80	00	29	A2	0F	BA	02	4E-4F	20	4E	41	4D	45	20	20
050	20	20	46	41	54	33	32	20-20	20	33	C9	8E	D1	BC	F4
060	7B	8E	C1	8E	D9	BD	00	7C-88	56	40	88	4E	02	8A	56
070	40	B4	41	BB	AA	55	CD	13-72	10	81	FB	55	AA	75	0A
080	F6	C1	01	74	05	FE	46	02-EB	2D	8A	56	40	B4	08	CD
090	13	73	05	B9	FF	FF	8A	F1-66	0F	B6	C6	40	66	0F	B6
0a0	D1	80	E2	3F	F7	E2	86	CD-C0	ED	06	41	66	0F	B7	C9
0b0	66	F7	E1	66	89	46	F8	83-7E	16	00	75	39	83	7E	2A
0c0	00	77	33	66	8B	46	1C	66-83	C0	0C	BB	00	80	B9	01
0d0	00	E8	2C	00	E9	A8	03	A1-F8	7D	80	C4	7C	8B	F0	AC
0e0	84	C0	74	17	3C	FF	74	09-B4	0E	BB	07	00	CD	10	EB
0f0	EE	A1	FA	7D	EB	E4	A1	7D-80	EB	DF	98	CD	16	CD	19
100	66	60	80	7E	02	00	0F	84-20	00	66	6A	00	66	50	06
110	53	66	68	10	00	01	00	B4-42	8A	56	40	8B	F4	CD	13
120	66	58	66	58	66	58	66	58-EB	33	66	3B	46	F8	72	03
130	F9	EB	2A	66	33	D2	66	0F-B7	4E	18	66	F7	F1	FE	C2
140	8A	CA	66	8B	D0	66	C1	EA-10	F7	76	1A	86	D6	8A	56
150	40	8A	E8	C0	E4	06	0A	CC-B8	01	02	CD	13	66	61	0F
160	82	74	FF	81	C3	00	02	66-40	49	75	94	C3	42	4F	4F
170	54	4D	47	52	20	20	20	20-00	00	00	00	00	00	00	00
180	00	00	00	00	00	00	00	00-00	00	00	00	00	00	00	00

Cursor pos = 0; log sec = 0; phy sec = 128

h) Click the folder **[root] (Root Folder**). What is the **logical sector number** that stores the **Root Folder**?

Answer: From the screenshot below, the logical sector number that stores the Root Folder is 8192

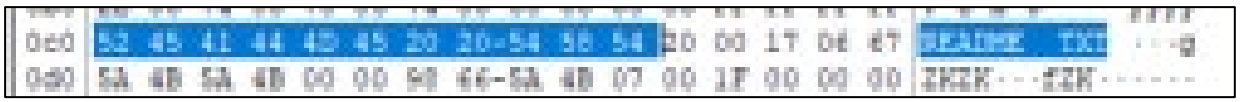


i) Examine the **hexadecimal values** in **Root Folder.**

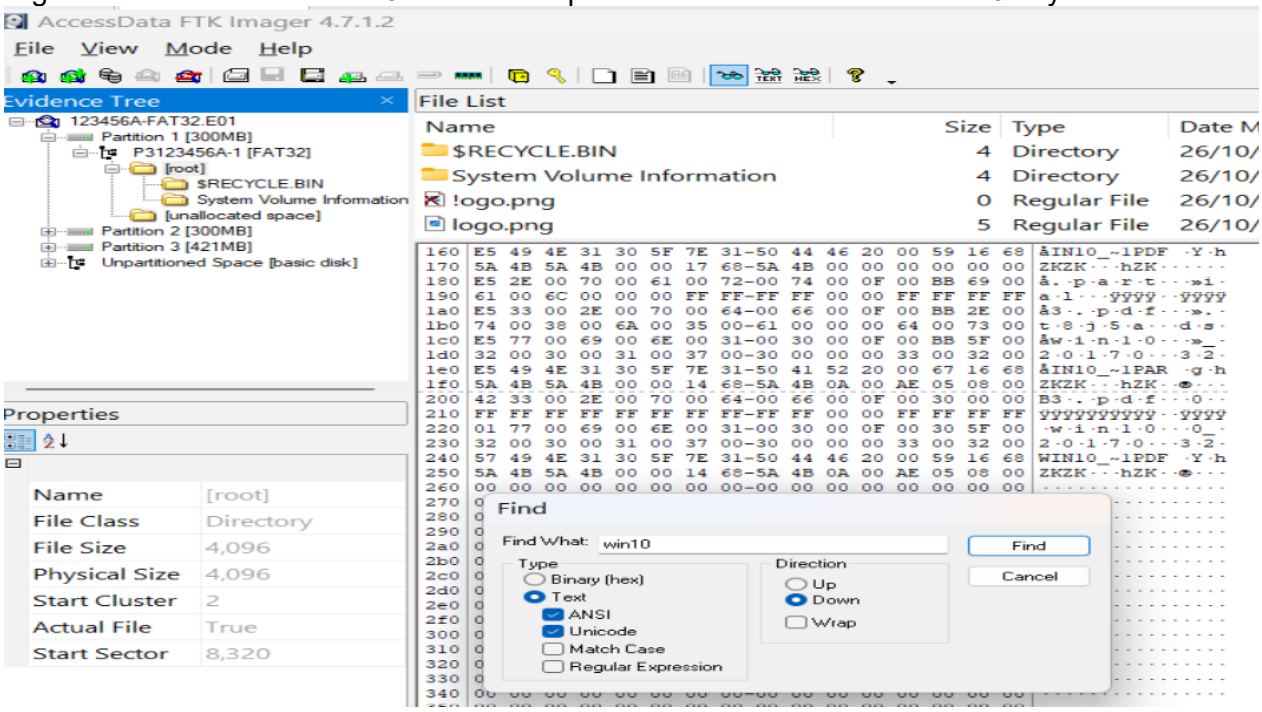
Browse to locate the file record for **win10_20170323.pdf** in **Root Folder**

What is the short filename (an 8.3 filename) of the file **win10_20170323.pdf** in **Root Folder**?
Capture the screen to show the file entry **win10_20170323.pdf** in **Root Folder**.

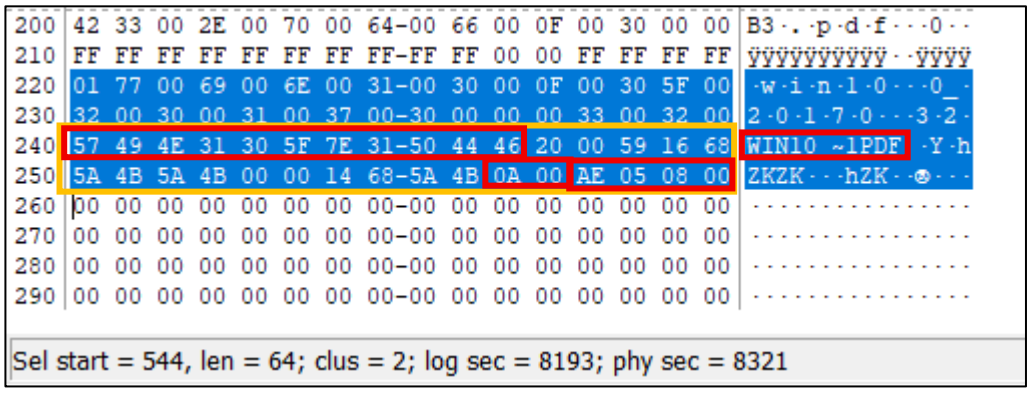
Below is a sample screenshot for the file entry for **Readme.txt**



Answer:
Right click->Find->Text “win10” ->with multiple “Find Next” to locate the last 32 bytes



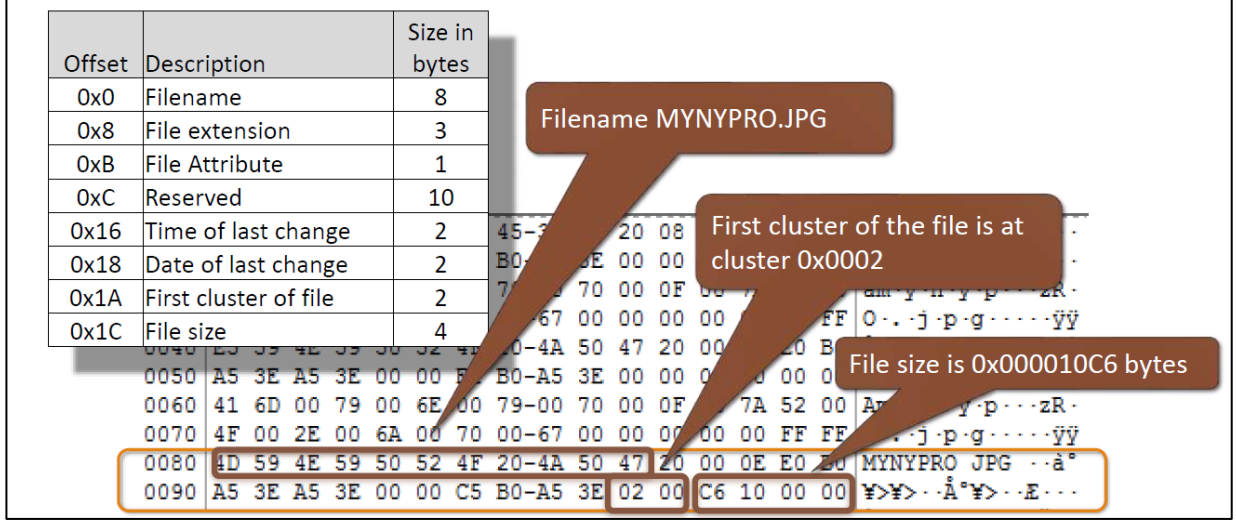
After browsing to find the file **win10_20170323.pdf** in Root Folder, 64-byte block of data is located.
The last 16 bytes store the short file name of **win10_20170323.pdf** which is **WIN10_~1PDF,** starting cluster and file size.



j) Find the size, the first cluster number and the last cluster number that store each of the following files. Put your answers in the following table.

Root Folder at Sector 19

Offset	Description	Size in bytes
0x0	Filename	8
0x8	File extension	3
0xB	File Attribute	1
0xC	Reserved	10
0x16	Time of last change	2
0x18	Date of last change	2
0x1A	First cluster of file	2
0x1C	File size	4



sample of FAT of a FAT32 file system is shown below

The FAT32 file system uses 32 bits per FAT entry, thus one entry spans four bytes in little-endian byte order. The four top bits of each entry are reserved for other purposes; they are cleared during formatting and should not be changed otherwise. They must be masked off before interpreting the entry as 28-bit cluster address.

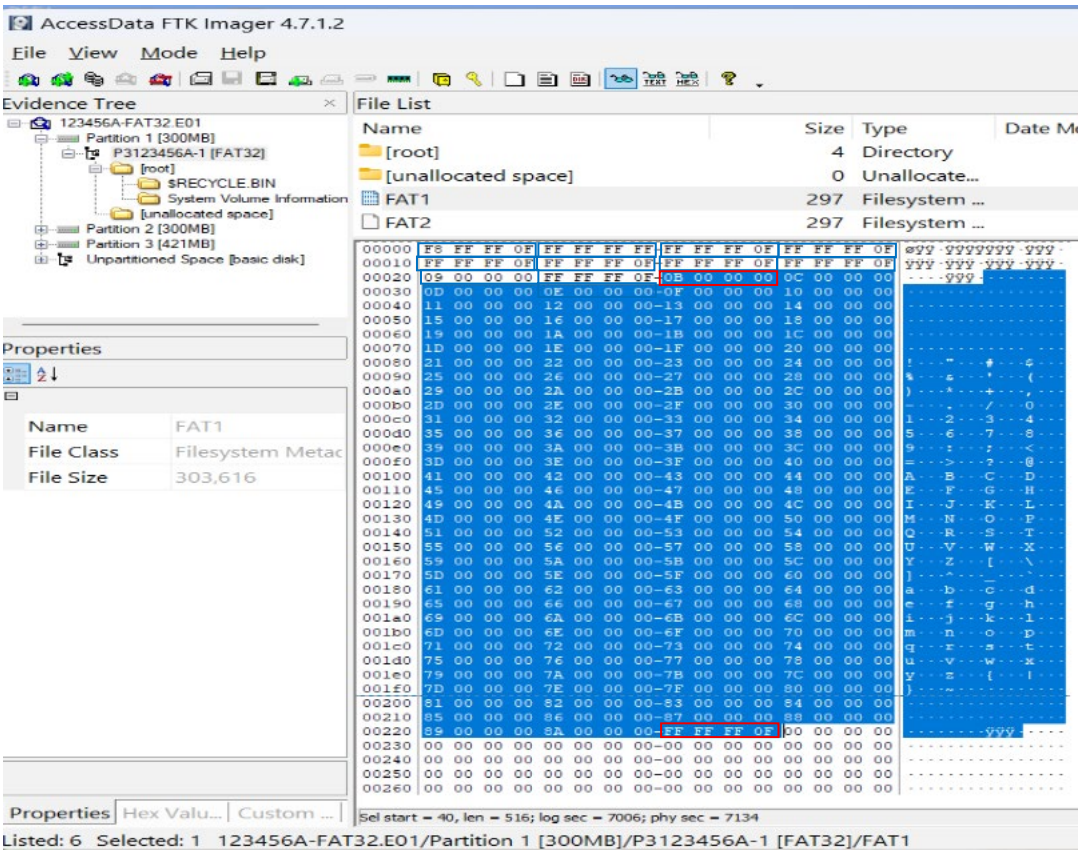
Example of FAT32 table start with several cluster chains

Offset	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
+0000	F0	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	04	00	00	00
+0010	05	00	00	00	06	00	00	00	07	00	00	00	08	00	00	00
+0020	FF	FF	FF	0F	0A	00	00	00	14	00	00	00	0C	00	00	00
+0030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
+0040	11	00	00	00	FF	FF	FF	0F	00	00	00	00	FF	FF	FF	0F
+0050	15	00	00	00	16	00	00	00	19	00	00	00	F7	FF	FF	0F
+0060	F7	FF	FF	0F	1A	00	00	00	FF	FF	FF	0F	00	00	00	00
+0070	00	00	00	00	F7	FF	FF	0F	00	00	00	00	00	00	00	00

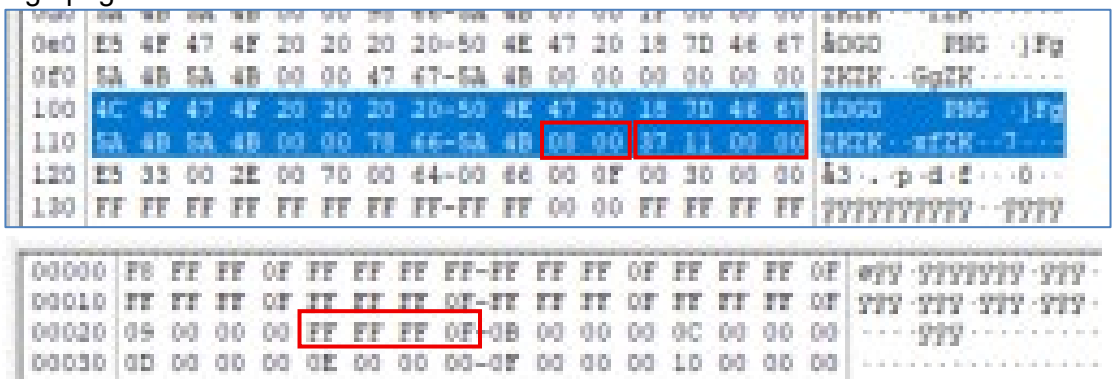
- First chain (1 cluster) for the root directory, pointed to by an entry in the FAT32 BPB (here: #2)
- Second chain (6 clusters) for a non-fragmented file (here: #3, #4, #5, #6, #7, #8)

Select FAT1

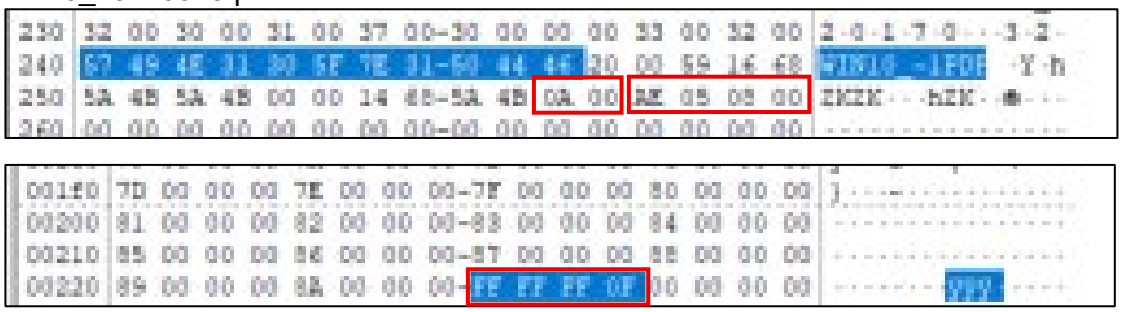
Given it is FAT32 system, each entry will occupy 32 bits (4 bytes),
Starting with 1st entry as 0, to locate start cluster at 0x0A (10), 4 bytes each will reach entry 0B 00 00 00
Following the cluster chain 0B->0C->0D…..->8A->reach the end of chain indicator FF FF FF 0F.
So the last cluster is 8A.



logo.png



win10_20170323.pdf



File Name	Size in bytes (decimal)	First Cluster (in hexadecimal)	Last Cluster (in hexadecimal)
Readme.txt	31	0x0007	0x00000007
logo.png	4,407(0x00001137)	0x0008	0x00000009
win10_20170323.pdf	525,742 (0x000805AE)	0x000A	0x0000008A